



Prenatal Diagnosis

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Objectives

By the end of this lecture, you should be able to:

1. Define prenatal diagnosis.
2. Differentiate between screening, diagnostic, invasive and non-invasive tests.
3. Discuss the different screening approaches.
4. Explain maternal and fetal screening
5. Explain the indications and utility of genetic testing.

Prenatal diagnosis

- Testing the mother or the fetus during the pregnancy to determine the risk of certain abnormalities
- To provide families with information to make informed choices during pregnancy



Goals of prenatal diagnosis

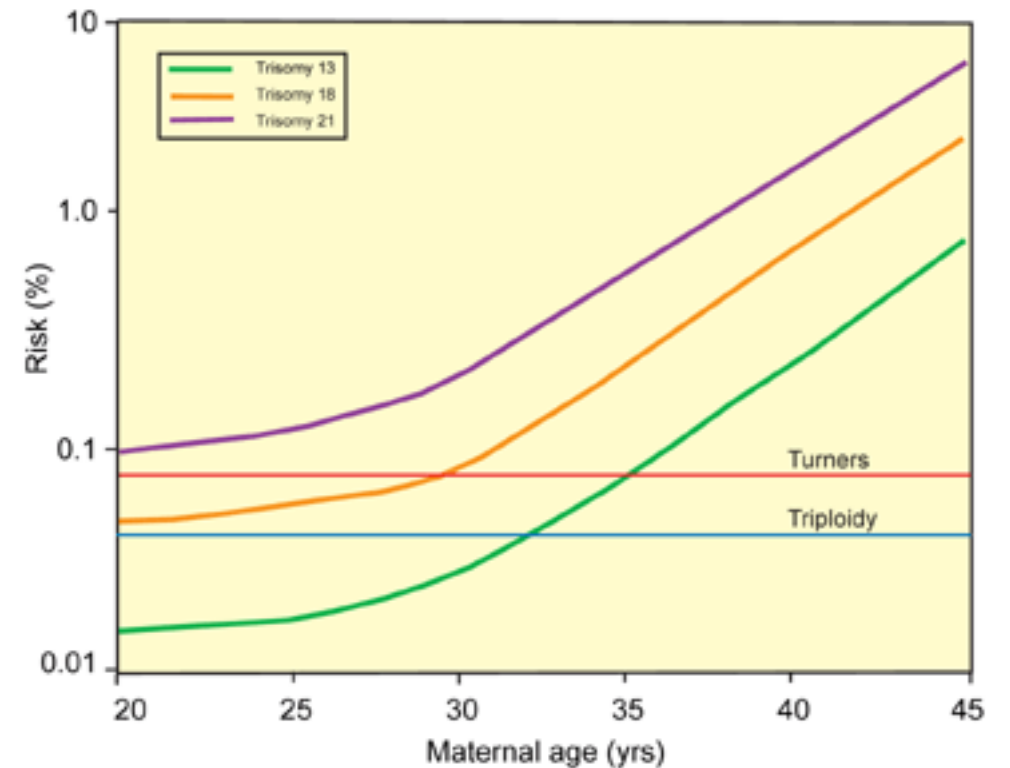
- Providing risk information to couples
 - General population vs high risk couples
 - **in the absence of such information, would not choose to begin a pregnancy**
 - Termination vs psychological preparation
 - **More likely to receive a normal then an abnormal result**
- Aid health-care professional to plan delivery, management, and care
 - **Congenital anomalies account for 20 to 25% of prenatal deaths.**

Indications for Prenatal Diagnosis

- Significant risk of disease:
 - Previous child or a family history of chromosomal abnormality
 - Family history of single gene disorder
 - Family history of congenital anomaly
- A serious condition to justify termination of pregnancy.
- Advanced maternal age
 - Chromosome 21 non-disjunction risk
 - $\sim 1/2000$ in a 25-year-old female
 - $\sim 1/200$ in a 35-year-old female
 - $\sim 1/20$ in a 45 year old female

Maternal age

Maternal Age	Incidence of Trisomy 21	Incidence of Trisomy 18	Incidence of Trisomy 13
15-19	1:1600	1:17000	1:33000
20-24	1:1400	1:14000	1:25000
25-29	1:1100	1:11000	1:20000
30-34	1:700	1:7100	1:4000
35-39	1:240	1:2400	1:4800
40-44	1:70	1:700	1:1600
45-49	1:20	1:650	1:1500



Screening vs Diagnostic

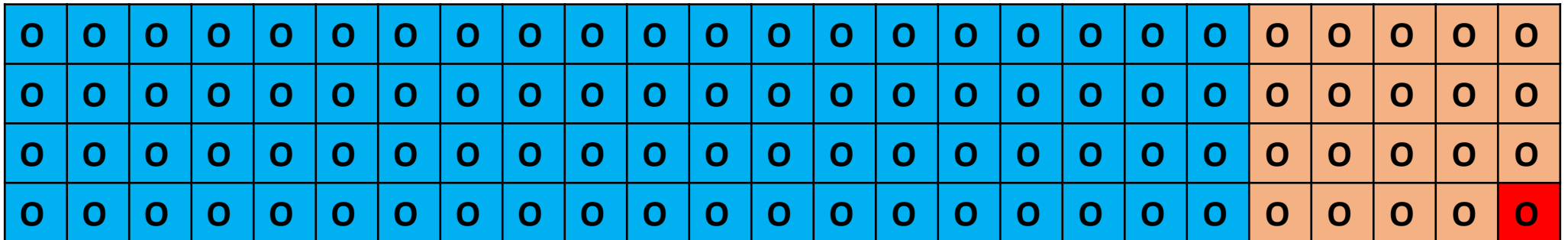
- Screening

- Used to identify individuals with a higher chance of a health concern; “POTENTIAL” health condition.
 - Has a higher risk of false positive

- Diagnostic

- A test used to establish the presence of, confirm, a diagnosis.
 - Usually more expensive or invasive

Example	
Population	100
+ve screening	20
+ve diagnosis	1



Methods Of Prenatal Diagnosis

Non-Invasive

- Ultrasonography Screening/Diagnostic
- Maternal blood (serum) Screening
- Fetal Cells in Maternal Blood Screening
- Fetal Echocardiography Screening/Diagnostic

Invasive Diagnostic

- Chorionic villus sampling
- Amniocentesis
- Percutaneous umbilical blood sampling
- Embryoscopy & fetoscopy



Non-invasive tests

Maternal Blood (Serum)



Serum Biomarkers

- Pregnancy induces physiological, hormonal & biochemical changes.
- When these deviate from the normal course, biochemical markers are altered
- Biochemical markers are used to assess fetal & maternal health
- Normal values change with gestational age

(Measured at the correct gestational age to enable accurate interpretation)

Serum Biomarkers

Help to diagnose and monitor

- **Maternal conditions**

- Gestational diabetes
- Pre-eclampsia
- Trophoblastic disease

- **Fetal abnormalities**

- Down's syndrome
- Neural tube defects

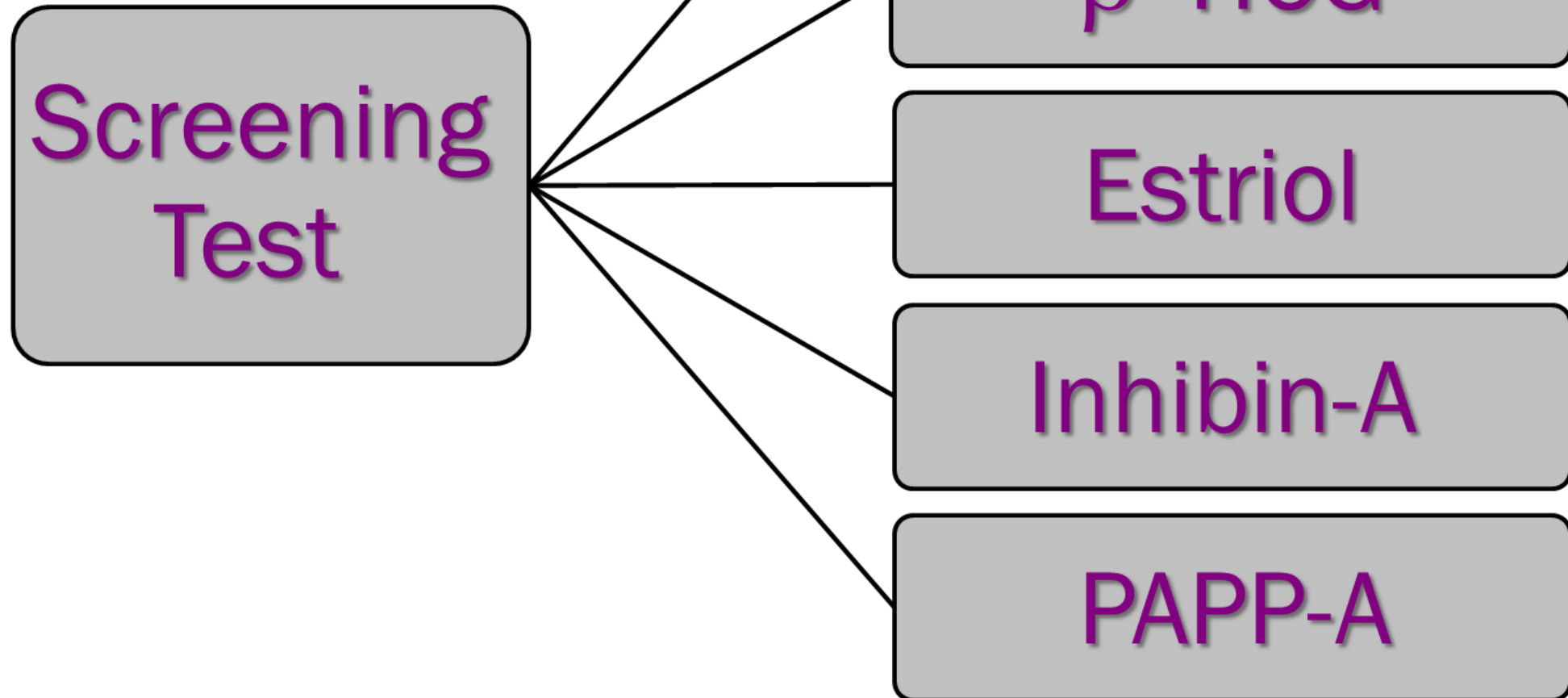


Biochemical Assessment of Fetal Health

- The major aim of fetal assessment is to ensure satisfactory growth in utero.
- Many factors cause fetal growth restriction:
 - Poor maternal nutritional state.
 - Placental insufficiency.
 - Fetal abnormality.



Maternal Serum Screening (MSS)



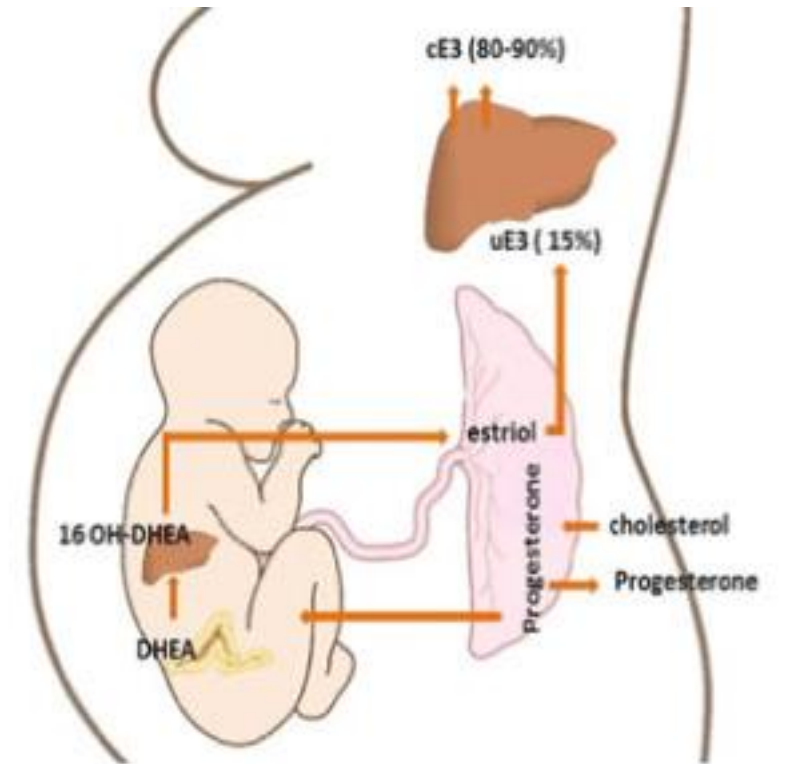
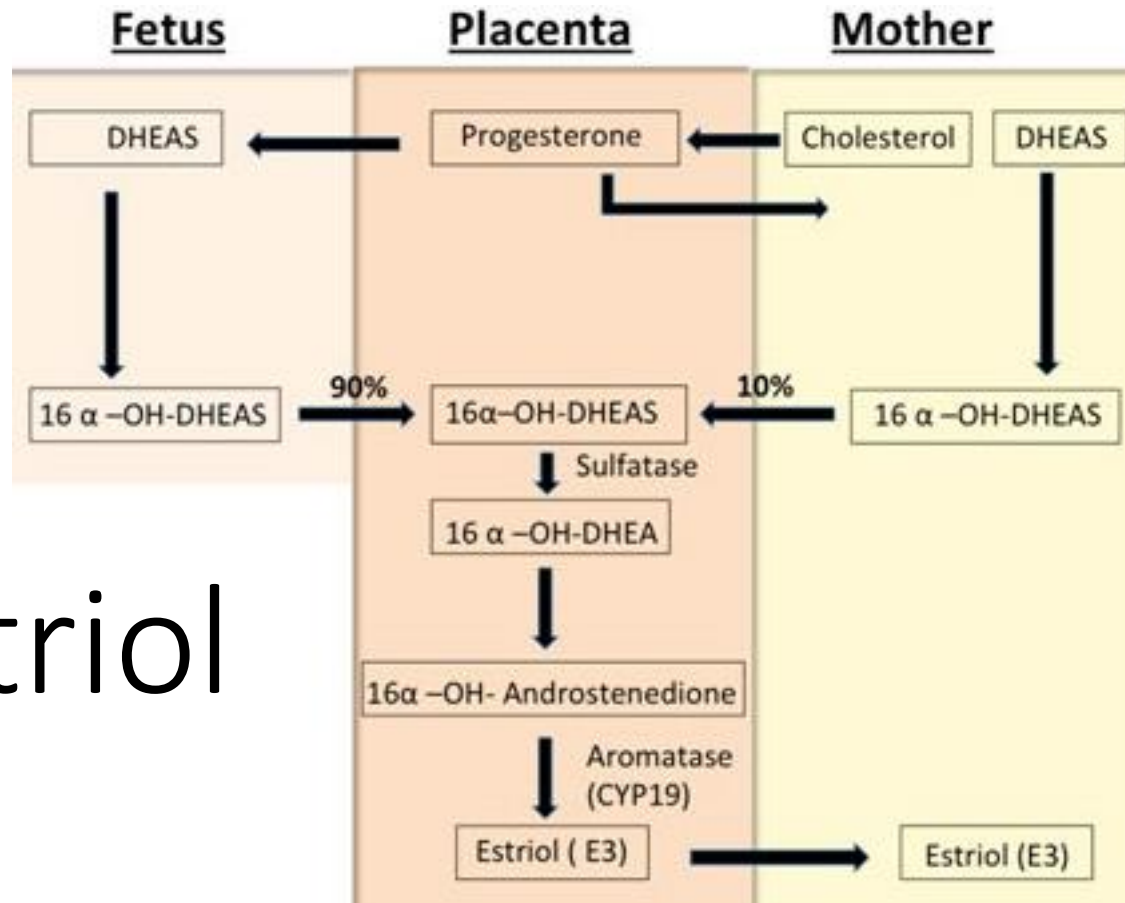
α -Fetoprotein (AFP)

- A fetal protein arising from the yolk sac and fetal liver.
- **Causes of Increased AFP**
 - Incorrect gestational age.
 - Threatened abortion
 - Multiple pregnancies
- **Fetal abnormality:**
 - Neural tube defects
 - Abdominal wall defect.
 - Skin defects
 - Renal anomalies

B-subunit of human chorionic gonadotropin (B-HCG)

- Commonly used as a test for pregnancy
- An increased B-HCG with a reduced AFP suggests a possibility of Down syndrome
- **Very high levels of HCG suggest trophoblastic disease**
- HCG can be used to follow up treatment for molar pregnancy to make sure that no residual trophoblastic disease persists.
Example: choriocarcinoma.

Estriol



Maternal Serum E3 (Esteriol)

- Serial E3 measurements in the 3rd trimester is a marker of general fetal status and well being.
- If E3 drops, then the fetus is threatened, and delivery may be necessary emergently.
- E3 is **reduced** in Down syndrome, adrenal hypoplasia, anencephaly or steroid hormone synthesis defects.

Inhibin-A

- Inhibin is secreted by the placenta and the corpus luteum.
- **Increased** inhibin-A is seen with **trisomy 21** and associated with a higher risk of preterm delivery.

Pregnancy-Associated Plasma Protein A (PAPP-A)

- Low PAPP-A in the 1st trimester is associated with fetal **chromosomal abnormalities, T13, 18 & 21.**
- In addition, also a predictor of adverse pregnancy outcomes
 - **Example: small for gestational age (SGA) or stillbirth.**
- Elevated PAPP-A level is seen in fetuses large for gestational age.



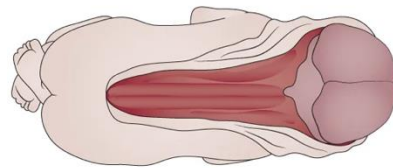
Conditions Detected with MSS



Neural Tube Defects

Neural Tube Defects

- Neural tube defects includes:
 - **Anencephaly** (failure of closure at the cranial end of the neural tube)
 - **Spina bifida** (failure of closure at the caudal end of the neural tube).



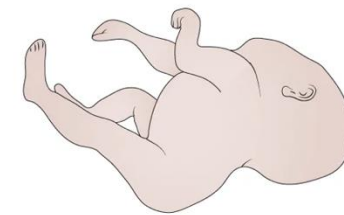
Craniorachischisis
Completely open brain and spinal cord



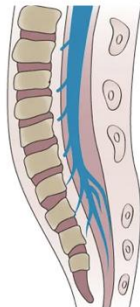
Anencephaly
Open brain and lack of skull vault



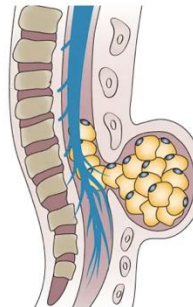
Encephalocele
Herniation of the meninges (and brain)



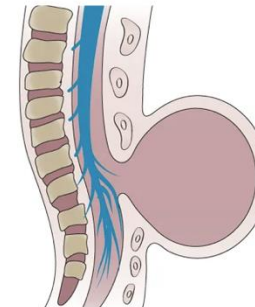
Iniencephaly
Occipital skull and spine defects with extreme retroflexion of the head



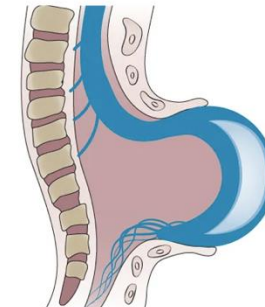
Spina bifida occulta
Closed asymptomatic NTD in which some of the vertebrae are not completely closed



Closed spinal dysraphism
Deficiency of at least two vertebral arches, here covered with a lipoma



Meningocele
Protrusion of the meninges (filled with CSF) through a defect in the skull or spine



Myelomeningocele
Open spinal cord (with a meningeal cyst)

Neural Tube Defects

- Incidence is about 1 to 2 births per 1000 in the United States.
- In NTDs, elevated AFP is present in both amniotic fluid and maternal serum.





Down Syndrome

Aneuploidies

- The pattern of MSS markers is used to assess the risk for Down syndrome and other aneuploidies

TABLE 17-6 Elevation and Depression of Parameters Used in First- and Second-Trimester Screening Tests

	First-Trimester Screen			Second-Trimester Screen			
	Nuchal Translucency	PAPP-A	Free β -hCG	uE ₃	AFP	hCG	Inhibin A
Trisomy 21	↑	↓	↑	↓	↓	↑	↑
Trisomy 18	↑	↓	↓	↓	↓	↓	—
Trisomy 13	↑	↓	↓	↓	↓	↓	—
Neural tube defect	—	—	—	—	↑↑	—	—

AFP, Alpha-fetoprotein; β -hCG, human chorionic gonadotropin β subunit; PAPP-A, pregnancy-associated plasma protein A; uE₃, unconjugated estriol.

Nuchal Translucency

- Space between back of the neck and overlying skin of the fetus
- Measured between 11 & 13 weeks.
- ↑ in ~70% of Down syndrome.



"Triple" or "Quadruple" Screen

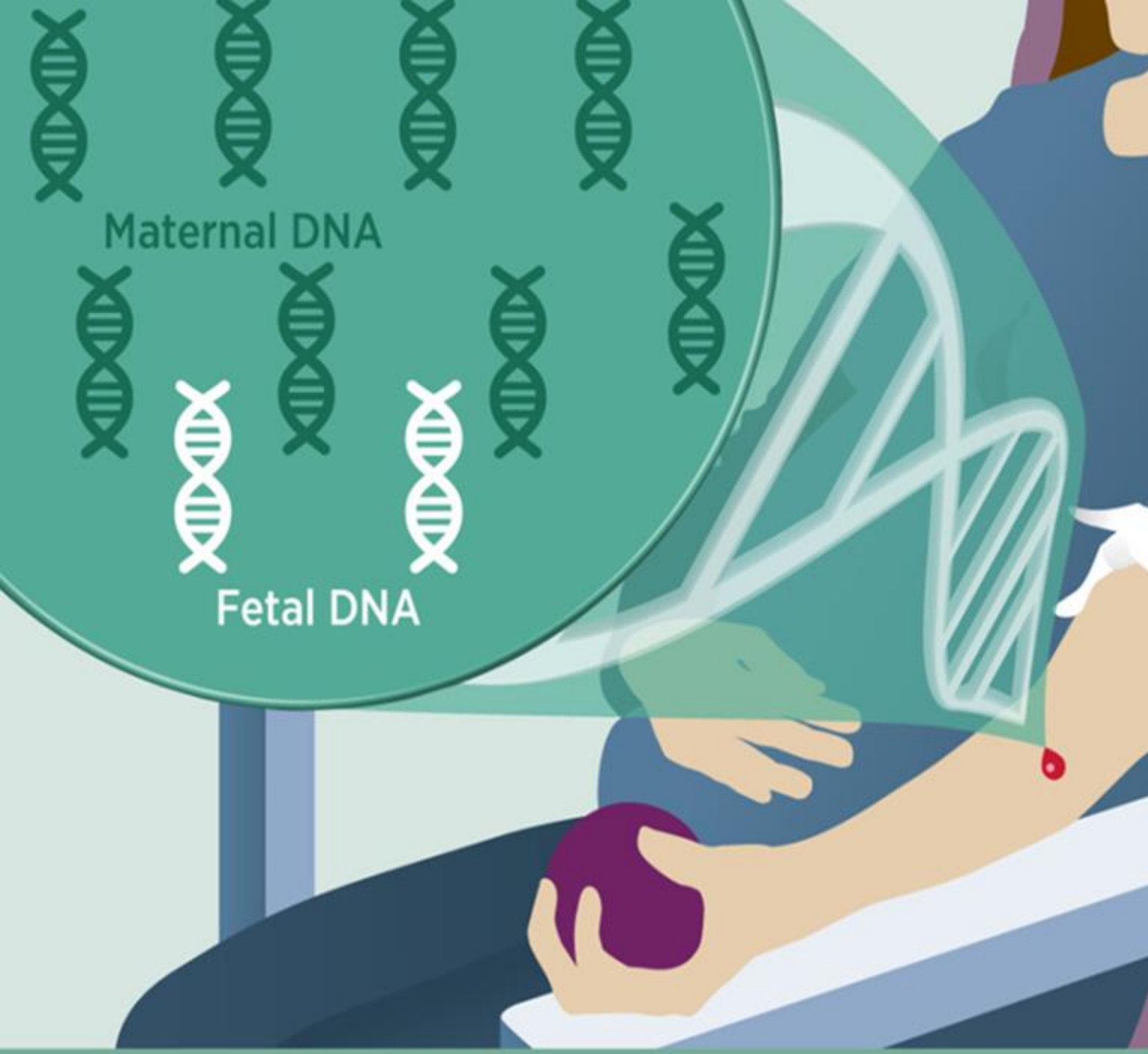
- Combining the maternal serum assays aids in increasing the sensitivity and specificity for detecting fetal abnormalities
 - The classic test “triple screen” **(first trimester screen)**
 - MSAFP, B-HCG & uE3.
 - The "quadruple screen" **(second trimester screen)**
 - adds inhibin-A (+ MSAFP, B-HCG & uE3)

Note:

Levels of these analytes change markedly during pregnancy

Interpretation depends greatly on knowing the correct gestational age, otherwise, results can be misinterpreted.

- Abnormal results **MUST BE** followed with direct karyotyping, by invasive test, to confirm or exclude the diagnosis
 - Carries a 0.5–1.0% risk of pregnancy loss



Fetal Cells in Maternal Blood

Non-invasive prenatal testing/screening (NIPT/S)

- Utilizes Cell free DNA (cfDNA)
 - Derived from the placental trophoblastic cells
 - Can be detected after 7+ weeks gestation
 - Undetectable within hours delivery

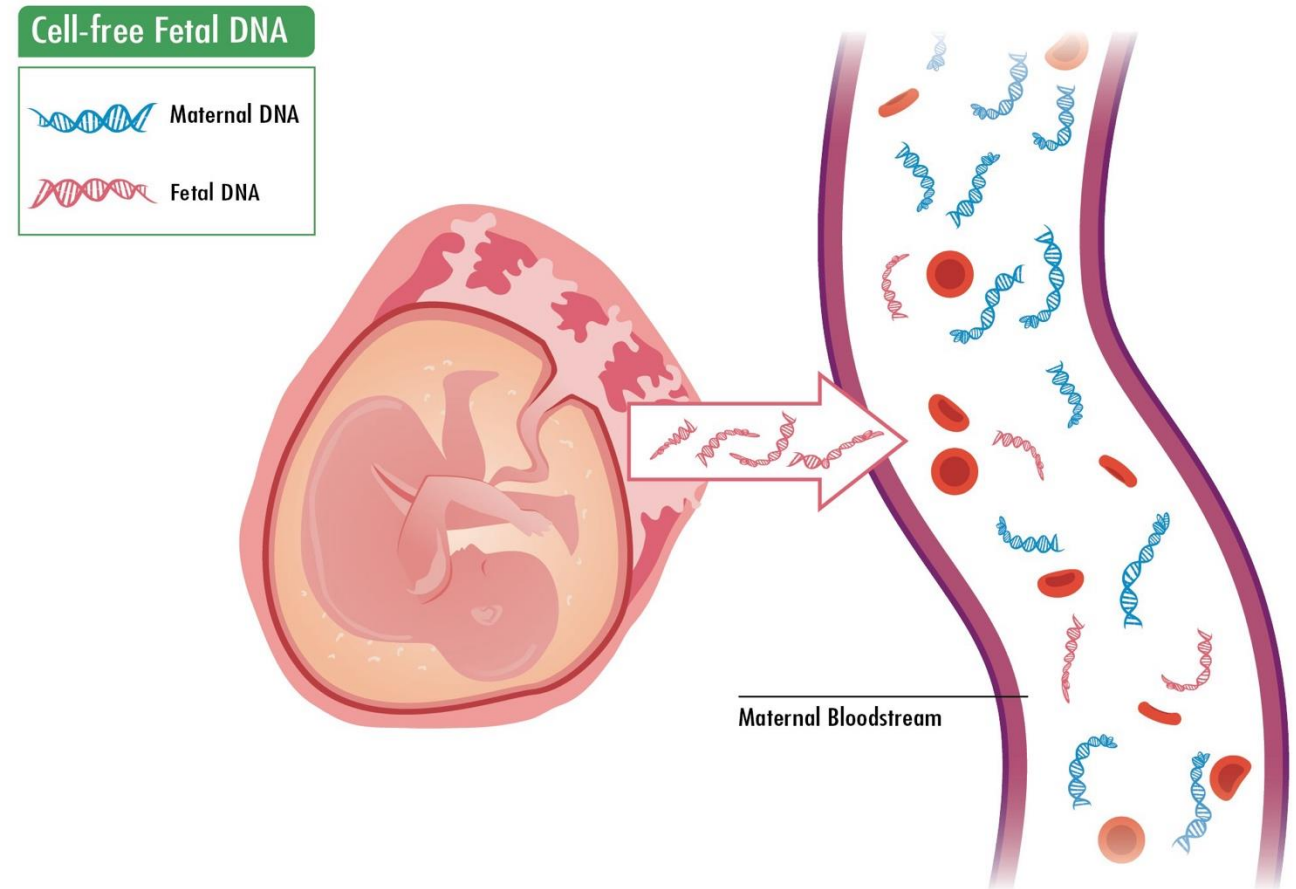


Table 20–1. Sensitivity and Specificity of NIPT for Common Aneuploidies

Aneuploidy	Sensitivity	Specificity
Trisomy 21	0.994	0.999
Trisomy 18	0.977	0.999
Trisomy 13	0.906	1.000
Monosomy X	0.929	0.999

Source: The meta-analysis of [Mackie et al. \(2017\)](#).

Table 20–2. Positive Predictive Values of NIPT Results at Different Maternal Ages

	PPV (%) at Maternal Ages			
Aneuploidy	25 Years	30 Years	35 Years	40 Years
Trisomy 13	7	10	21	50
Trisomy 18	15	21	39	69
Trisomy 21	51	61	79	93

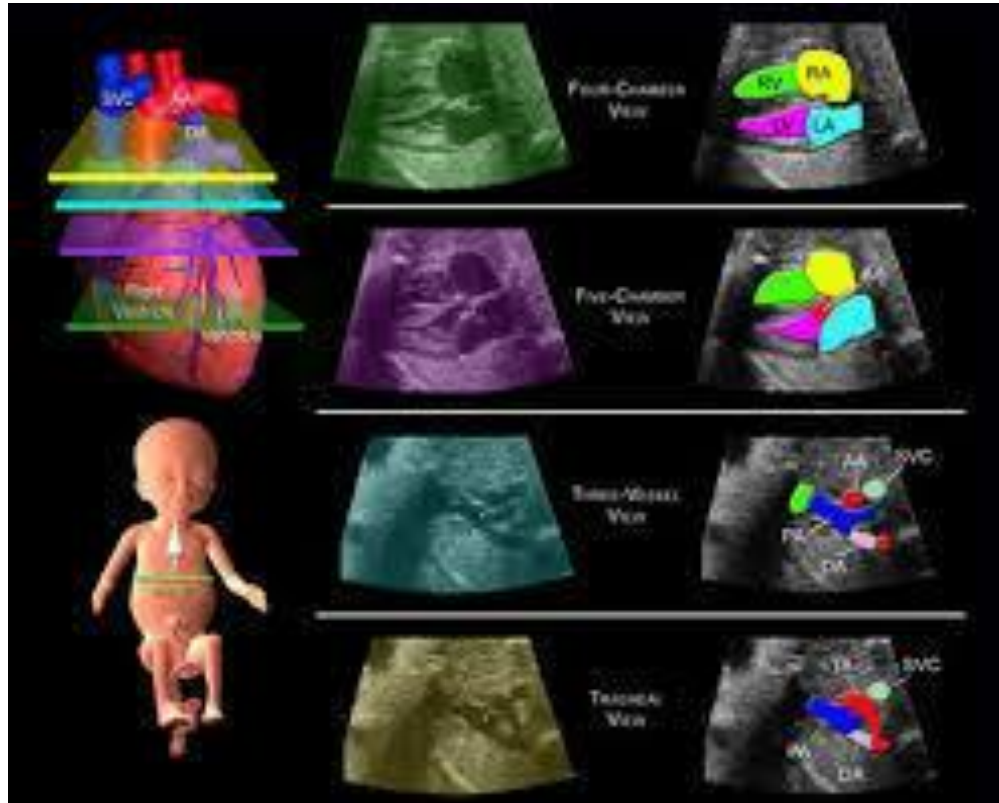
Discordant NIPT

- Analytical issues
 - Failure of extraction or amplification steps
 - Very low fetal fraction
- Placental mosaicism
 - Can also give a false negative result
- Twins
 - Resolving second gestational sac/vanishing twin/demise
 - Second twin DNA detected >8week after it's demise
- CNVs or mosaicism in mother
 - SCA more common → 8.6% of positive NIPT for SCA are due to maternal mosaicism
- Maternal malignancies

Ultrasonography



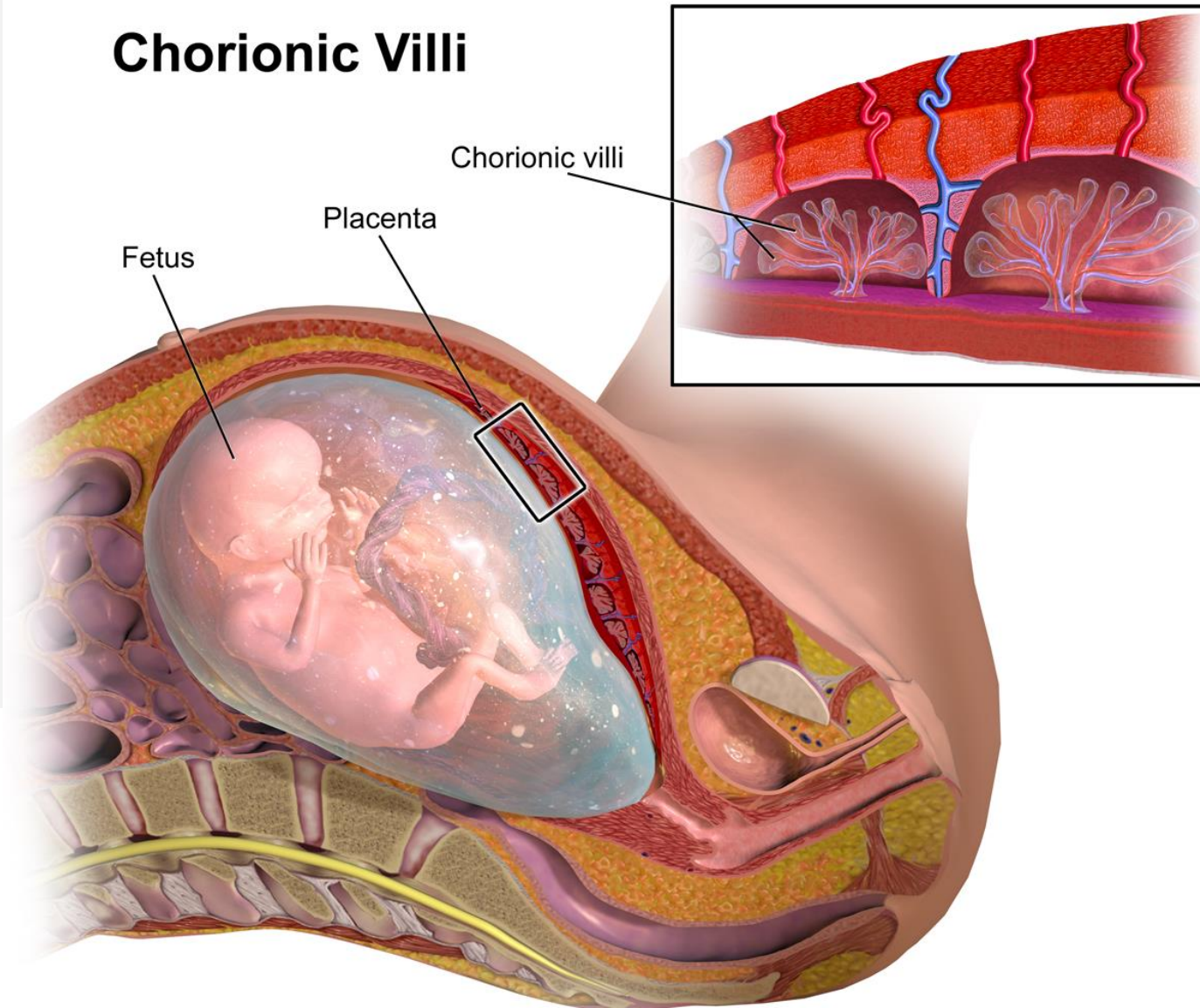
Fetal Echocardiography





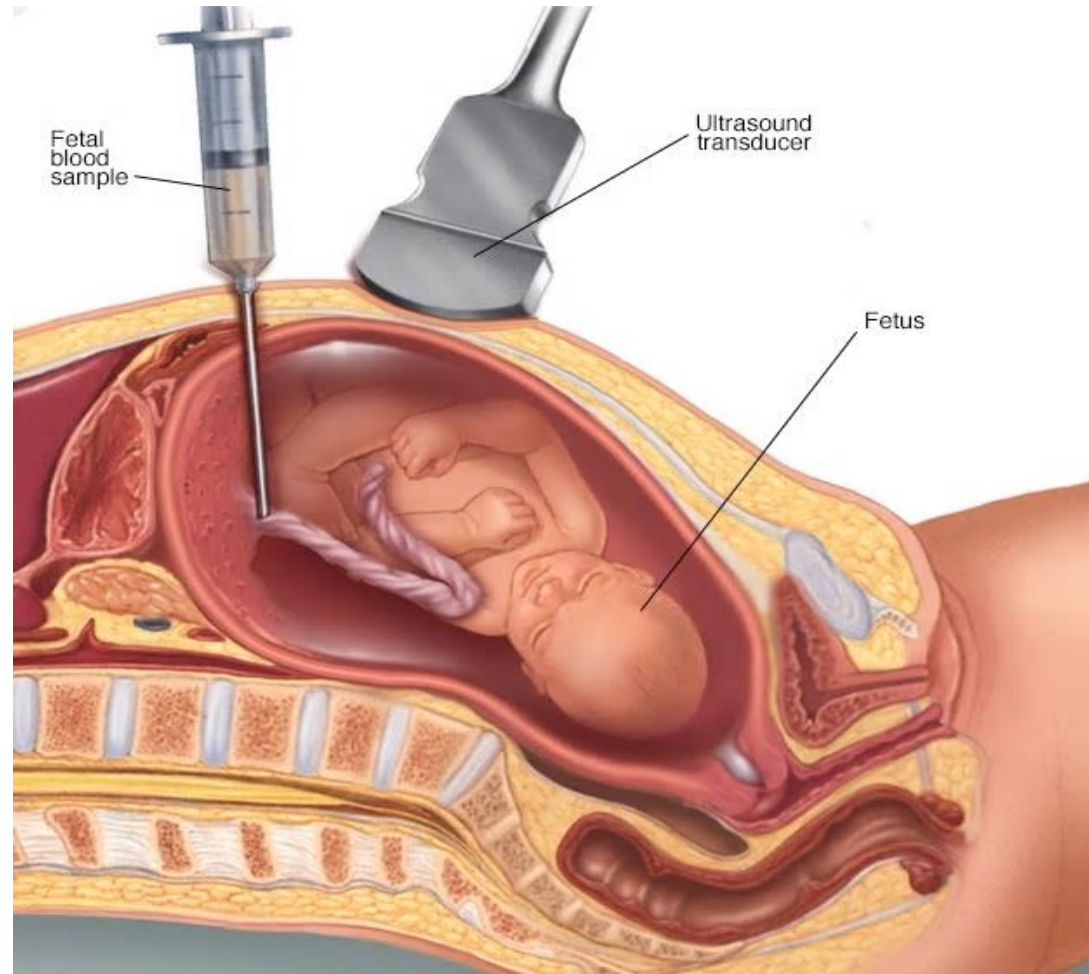
Invasive Test

Chorionic Villi

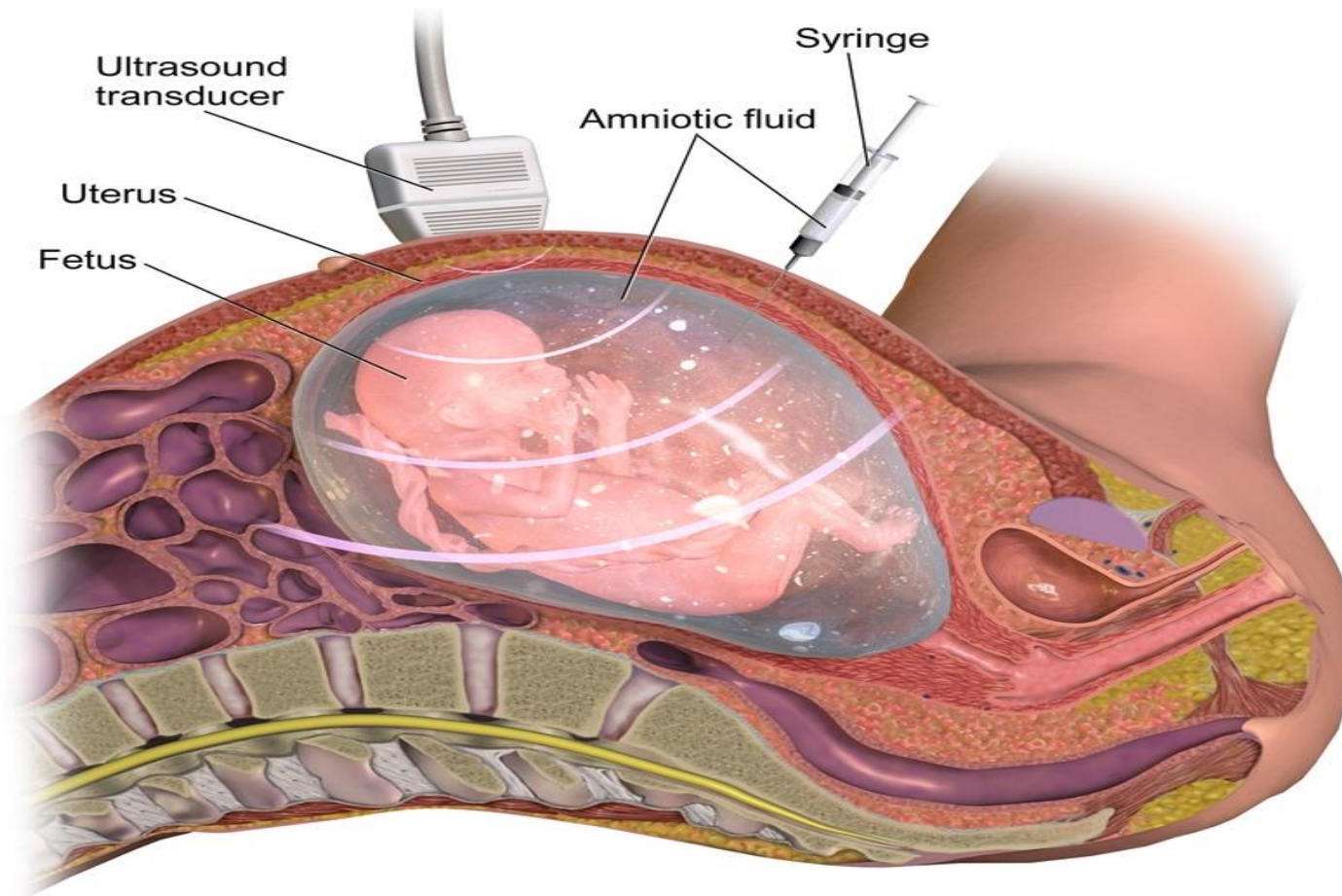


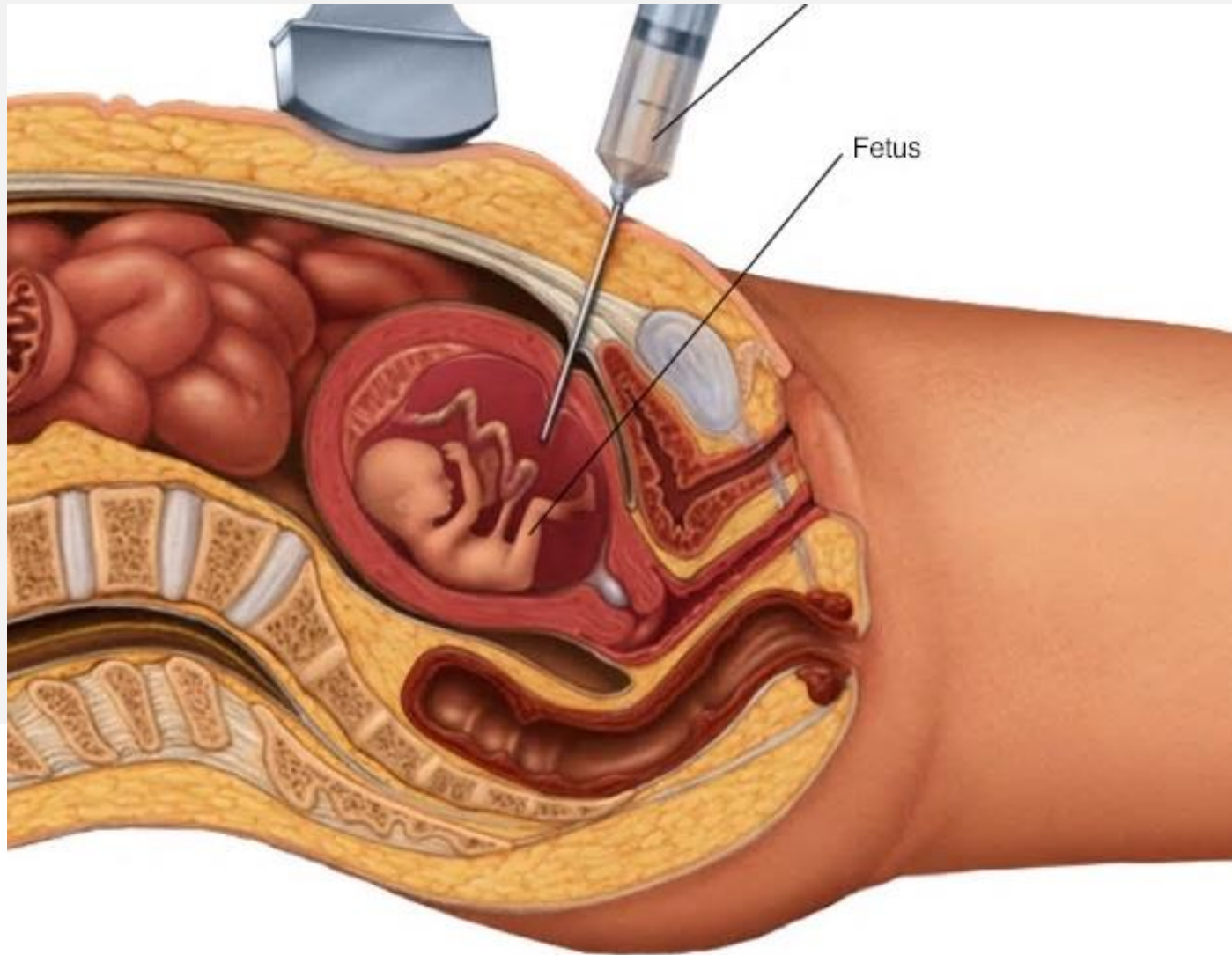
Chorionic Villus Sampling

Percutaneous Umbilical Blood Sample (Cordocentesis)



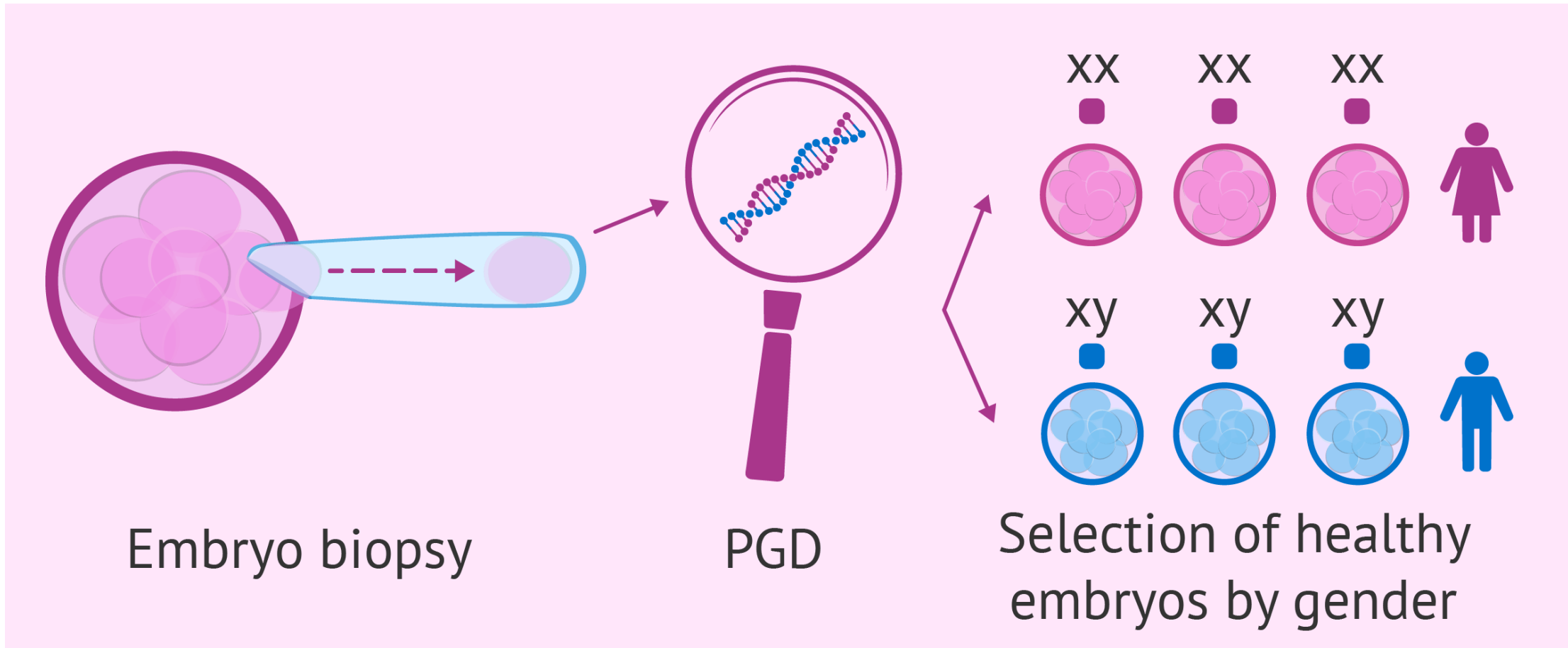
Chorionic Villus Sampling





Amniocentesis

Preimplantation Genetic Diagnosis (PGD)



Special Problems In Prenatal Diagnosis

- Failure to obtain a sample or culture failure
- An ambiguous chromosome result
- An unexpected chromosome result
- Ultrasonographic “soft marker”



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Thank you